

ROLAND: Robust Landing of UAV on Moving Platform using Object Detection and UWB based Extended Kalman Filter

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Abstract: Unmanned Aerial Vehicles (UAVs) have been intensively used in various fields thanks to their excellent maneuverability. However, the short operation time of the UAV is limiting the further utilization of the UAV. The limitation of the UAV can be overcome by landing on ground vehicles and charging the battery on them. Therefore, diverse researches have been done on robust landing on ground platforms. Unfortunately, the existing autonomous landing methods have such limitations that a) special tag or marker should be attached on the landing site; b) visibility of the landing site must be secured; c) platform should be static; In this paper, to robustly estimate the relative pose of the moving platform and successfully land on it, we propose a novel robust landing system of the UAV. A neural network based object detection is used to recognize the landing site without a special tag or marker, and an Ultra-wideband (UWB) sensor is adopted to compensate for the limited field of view of the camera. The Extended Kalman Filter (EKF) for estimating the relative position of the moving platform by fusing the information obtained from various sensors is developed. Additionally, a robust autonomous landing controller is also designed. The performance of the proposed method is verified by the simulation of the UAV and the moving platform.

Keywords: Unmanned Aerial Vehicles, Autonomous Landing, Relative Pose Estimation, Extended Kalman Filter, Object Detection, Ultra-wideband

1. INTRODUCTION

Unmanned Aerial Vehicles (UAVs) have been effectively used in various fields such as transportation, structural monitoring, and exploration, based on their relatively simple structure, excellent maneuverability, and hovering ability [1]-[5]. However, compared to other unmanned vehicles, relatively short operating time or limited payload of the UAV have been blocking the further utilization of UAVs. The limitation of UAVs can be overcome by landing on an Unmanned Ground Vehicle (UGV) or Unmanned Surface Vehicle (USV) and charging the battery or replacing the sensors on them. Therefore, diverse researches have been done on precise and robust landing on movable platforms. Thanks to the advantages of providing a lot of information considering its small size and light weight, cameras have been mainly used to recognize the landing site and guide the UAVs [6]-[10]. Unfortunately, the autonomous landing method using only visual information has a limitation in that visibility of the landing site must be secured throughout the whole landing process because it cannot track the landing site when it is out of the camera's Field of View (FoV) or is obscured by obstacles.

To solve these problems, in this paper, we propose "ROLAND: ROBust LANDING of UAV on Moving Platform Using Object Detection and UWB based Extended Kalman Filter," a method for robust autonomous landing on a moving platform. An RGB-D camera and ob-

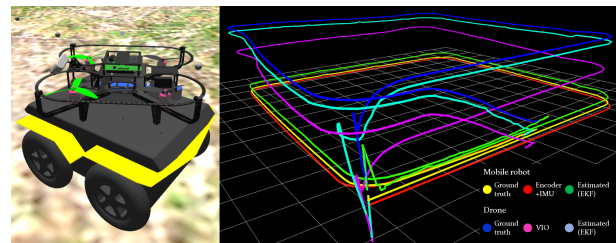


Fig. 1. An instance of the autonomous landing simulation. (Left) The UAV successfully landed on the moving platform. (Right) Trajectories including ground truth, encoder data, Visual-Inertial Odometry result, and estimated relative position by the designed EKF.

ject detection [17] are used to recognize the landing site and estimate the location of the moving platform without a special tag or marker, and an Ultra-wideband (UWB) sensor is adopted to compensate for the limited FoV of the camera. The Extended Kalman Filter (EKF) is developed to estimate the relative location of the landing site precisely and reliably by fusing the various sensors' measurements including the encoder, Visual-Inertial Odometry (VIO), object detection, and UWB. Moreover, a robust autonomous landing controller to land the UAV on a moving platform is also designed. Finally, the robust performance of the proposed autonomous landing system, ROLAND, is validated through the simulation using Gazebo [18].

2. RELATED WORKS

A wide range of research on landing an UAV at a specific location using visual information has been conducted. First, the method of landing through image-based visual servoing by recognizing the fiducial marker attached on the center of the landing pad was studied by D. Lee *et al.* [6]. In the study, it was shown that the quadrotor could land not only on the fixed landing pad but also on the landing pad which is moving at a constant speed through the proposed method. However, there are limitations that the specific fiducial marker must be attached to the landing site, and the marker must be located within the FoV of the UAV's downward-facing camera throughout the whole landing process.

For a more precise and robust landing, sensor fusion techniques have been introduced with multi-sensor systems, including cameras and IMUs. S. Yang *et al.* developed an autonomous landing system where visual measurement and IMU data were fused using Square Root Unscented Kalman Filter to estimate the location of the landing site [8]. However, a particular marker must be attached to the landing site, and it was not shown that the landing system is feasible when the landing site is not static. In [10], to land a drone on a moving target even when visual information is not reliable, the target's dynamic modeling was modeled to predict the target's position through the EKF. However, since it was assumed that the target is moving in a straight line at a constant velocity, landing on a target, which accelerates or changes the direction of motion, cannot be guaranteed.

3. METHODOLOGY

ROLAND is composed of an EKF that estimates the relative position between the UAV and the target and an autonomous landing controller, as shown in Fig. 2. The state of EKF is propagated using the Visual-Inertial Odometry (VIO) information of the UAV and the encoder value of the target, and the state is corrected and updated using object detection and UWB sensor. Finally, using the VIO information and the relative position of the target estimated from the EKF, the autonomous landing controller would land the UAV on the moving target.

3.1 Filter Design and Propagation

The proposed EKF estimates the position of the moving platform in the perspective of the UAV as below:

$$\mathbf{Y}_{3 \times 1} = [(\mathbf{p}_{\text{target}}^{\text{uav}})^T_{3 \times 1}]^T. \quad (1)$$

To obtain $\mathbf{Y}$, the indirect state variable based on the world-fixed coordinate system is defined as follows:

$$\mathbf{X}_{6 \times 1} = [(\mathbf{p}_{\text{uav}}^{\text{w}})^T_{3 \times 1} \quad (\mathbf{p}_{\text{target}}^{\text{w}})^T_{3 \times 1}]^T \quad (2)$$

where $\mathbf{p}_{\text{uav}}^{\text{w}}$ and $\mathbf{p}_{\text{target}}^{\text{w}}$ denote the position vectors of the UAV and the moving platform in the world frame, respectively. In the filter propagation stage, the state variable is

updated using the encoder value of the moving platform and the VIO result of the UAV as follows:

$$\begin{aligned} \mathbf{X}_k^- &= \mathbf{A} \hat{\mathbf{X}}_{k-1} + \mathbf{B}u \\ &= \hat{\mathbf{X}}_{k-1} + [(\Delta \mathbf{p}_{\text{uav}})^T \quad (\Delta \mathbf{p}_{\text{target}})^T]^T \end{aligned} \quad (3)$$

where $\mathbf{X}_k^-$ is the prediction of the state variable at the k -th time step and $\hat{\mathbf{X}}_{k-1}$ is the estimation of the state variable at the $k-1$ -th time step. Here, $\mathbf{A}$ denotes the state transition matrix, which equals to $\mathbf{I}_{6 \times 6}$. $\Delta \mathbf{p}_{\text{uav}}$ is the position change of the UAV obtained through VIO, and $\Delta \mathbf{p}_{\text{target}}$ is the position change of the moving platform measured by the encoder. Note that subscripts _{uav}, _{target} denote VIO and encoder fixed frames, respectively. By using encoder data, the position of the moving platform can be predicted even when the moving platform is not in the FoV of the camera mounted on the UAV. However, the position predicted through the encoder has the scale error and drift, which may vary depending on the surface condition of the operating area [16]. Similarly, the position of the UAV can be estimated using VIO, but there is also scale error, and drift can be accumulated depending on the flying trajectories [13]. The drift and error in prediction are corrected in further steps.

The covariance matrix $\mathbf{P}_k^-$ can be predicted as follows with the covariance of the previous time step $\mathbf{P}_{k-1}$:

$$\begin{aligned} \mathbf{P}_k^- &= \mathbf{A} \mathbf{P}_{k-1} \mathbf{A}^T + \mathbf{Q} \\ &= \mathbf{P}_{k-1} + \mathbf{Q} \end{aligned} \quad (4)$$

$$\mathbf{Q}_{6 \times 6} = \begin{bmatrix} \sigma_v^2 \Delta t_v \mathbf{I}_{3 \times 3} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \sigma_E^2 \Delta t_E \mathbf{I}_{3 \times 3} \end{bmatrix} \quad (5)$$

where σ_v^2 and σ_E^2 denote the variances of the VIO and encoder errors that occur every unit time, respectively. The irregular prediction due to the communication delay of the encoder and the VIO calculation are also considered with Δt in the process noise covariance matrix $\mathbf{Q}$. Subscripts _v and _E denote the encoder and VIO, respectively.

3.2 Filter Update with UWB

In the filter update step, the estimation of the state variable is updated through the UWB sensor and object detection. UWB sensor can measure the linear distance between the tag attached to the UAV and the anchor attached to the moving platform. The relationship between the state variable $\mathbf{X}$ and UWB measurement $\mathbf{Z}^u = d$ can be expressed as follows:

$$\mathbf{Z}^u = h^u(\mathbf{X}) = \|\mathbf{p}_{\text{uav}}^{\text{w}} - \mathbf{p}_{\text{target}}^{\text{w}}\| = d \quad (6)$$

where h^u denotes the nonlinear measurement function or Euclidean distance here. Thus, the state variable $\hat{\mathbf{X}}_k$ is updated with $\mathbf{Z}^u = d$ as follows:

$$\hat{\mathbf{X}}_k = \mathbf{X}_k^- + \mathbf{K}_k^u (\mathbf{Z}_k^u - \mathbf{H}^u \mathbf{X}_k^-) \quad (7)$$

$$6 \times 1 \mathbf{K}_k^u = \mathbf{P}_k^- (\mathbf{H}^u)^T (\mathbf{H}^u \mathbf{P}_k^- (\mathbf{H}^u)^T + \mathbf{R}_k^u)^{-1} \quad (8)$$

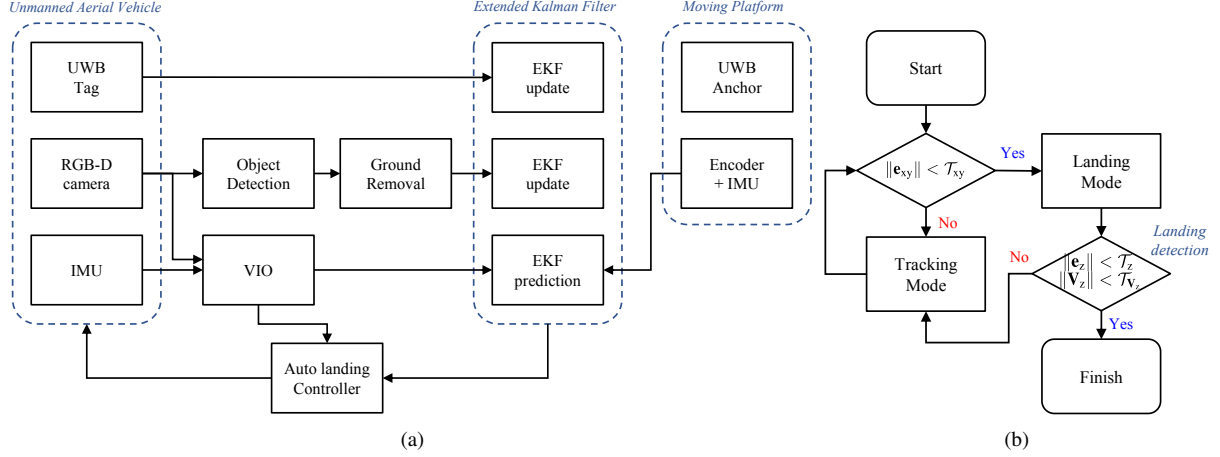


Fig. 2. The overall structures of the proposed systems, ROLAND. (a) The developed EKF. (b) The designed autonomous landing controller.

$$\mathbf{H}^u = \begin{bmatrix} \frac{\partial h^u(\mathbf{X})}{\partial \mathbf{p}_{uav}^w} & \frac{\partial h^u(\mathbf{X})}{\partial \mathbf{p}_{target}^w} \end{bmatrix}_{1 \times 6}, \mathbf{R}_k^u \propto d \quad (9)$$

where $\mathbf{K}_k^u$ is Kalman gain of filter update with UWB and $\mathbf{H}^u$ is the observation matrix of UWB sensor, which is a linearized result of h^u with Jacobian. $\mathbf{R}_k^u$ is the observation noise of UWB and is set as proportional to the distance d considering its characteristic [14]. The covariance $\mathbf{P}_k$ is updated with $\mathbf{P}_k^-$ and $\mathbf{K}_k^u$ as follows:

$$\mathbf{P}_k = (\mathbf{I}_{6 \times 6} - \mathbf{K}_k^u \mathbf{H}^u) \mathbf{P}_k^-. \quad (10)$$

3.3 Filter Update with Object Detection and Ground Removal

YOLOv4-tiny-3l network [17] is selected for detecting the moving platform accurately in real-time. A box bounding the moving platform is obtained by inferring the network and choosing the best result with the highest confidence, greater than the threshold $\mathcal{T}_{YOLO}$.

The pointcloud of the target inside the box is obtained from the depth of the RGB-D camera. However, since the detected result is in the form of a rectangular rather than an exact outline of the object, unnecessary ground points are included in the pointcloud, which may degrade the accuracy of its center point. Therefore, to estimate the exact location of the moving platform, the ground points should be removed properly. Assuming the points outside the box are not those of the moving platform, the points of the ground can be randomly sampled by selecting N points outside the box. Supposing the ground is flat enough, all ground points satisfy the equation of the plane as follows:

$$ax + by + cz = 1. \quad (11)$$

The constants a , b , and c can be obtained by solving the linear system equation as follows using Singular Value Decomposition (SVD):

$$\begin{bmatrix} x_1 & y_1 & z_1 \\ \vdots & \vdots & \vdots \\ x_N & y_N & z_N \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}. \quad (12)$$

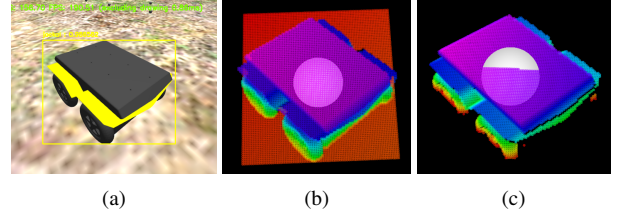


Fig. 3. The visual description of object detection and ground removal processes. (a) Detected moving platform. Yellow rectangular represents the bounding box. (b) Pointcloud within the bounding box. (c) Pointcloud after ground removal. The colors represent the corresponding height of data, while the white spheres are the center points both in (b) and (c).

By letting $\mathbb{P} = \{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_k, \dots, \mathbf{p}_N\}$ as the set of N points $\mathbf{p}$ in the bounding box, the set $\mathbb{Q}$ of pure points of the moving platform can be obtained as follows:

$$\mathbb{Q} = \mathbb{P} - \mathbb{G} \quad (13)$$

$$\mathbb{G} = \left\{ \mathbf{p}_k \in \mathbb{P} \mid \frac{|ax_k + by_k + cz_k - 1|}{\sqrt{a^2 + b^2 + c^2}} < \mathcal{T}_{\text{ground}} \right\} \quad (14)$$

where $\mathbb{G}$ is the set of points that are determined as unnecessary ground points among the pointcloud within the bounding box. $\mathcal{T}_{\text{ground}}$ denotes the distance tolerance from the ground plane to a point, which is used to classify the point as laid on the ground. For better understanding, visual description of the object detection and the ground removal are shown in Fig. 3. Afterward, the center point of $\mathbb{Q}$, $\mathbf{p}_{\text{target}}^c$, is used to update the filter, which should be transformed from the camera-fixed frame into the world-fixed frame as follows:

$$\begin{bmatrix} \mathbf{Z}_k^c \\ 1 \end{bmatrix} = \begin{bmatrix} {}^w\mathbf{R}_{uav} & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} {}^{uav}\mathbf{T}_c \begin{bmatrix} \mathbf{p}_{\text{target}}^c \\ 1 \end{bmatrix} \quad (15)$$

where ${}^w\mathbf{R}_{uav}$, ${}^{uav}\mathbf{T}_c$ represent the rotational matrix from the world-fixed to VIO-fixed frame and the SE(3) transformation matrix from the VIO-fixed to camera frame, respectively. The relationship between the state $\mathbf{X}$ and

the object detection measurement $\mathbf{Z}^c$ can be expressed by the observation model $\mathbf{H}^c$ as follows:

$$\mathbf{Z}^c = \mathbf{H}^c \mathbf{X} = \mathbf{p}_{\text{target}}^w - \mathbf{p}_{\text{uav}}^w \quad (16)$$

$$\mathbf{H}^c = \begin{bmatrix} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{bmatrix}. \quad (17)$$

The estimation $\hat{\mathbf{X}}_k$ is updated from the prediction $\mathbf{X}_k^-$ as follows:

$$\hat{\mathbf{X}}_k = \mathbf{X}_k^- + \mathbf{K}_k^c (\mathbf{Z}_k^c - \mathbf{H}^c \mathbf{X}_k^-). \quad (18)$$

$$6 \times 3 \mathbf{K}_k^c = \mathbf{P}_k^- (\mathbf{H}^c)^\top (\mathbf{H}^c \mathbf{P}_k^- (\mathbf{H}^c)^\top + \mathbf{R}_k^c)^{-1} \quad (19)$$

where $\mathbf{K}_k^c$ is Kalman gain of object detection. The observation noise of object detection $\mathbf{R}_k^c$ is set as proportional to the depth information noise of the RGB-D camera and the reciprocal of the confidence of the detected object from YOLOv4-tiny-3l. The covariance $\mathbf{P}_k$ of the filter is updated with the prediction $\mathbf{P}_k^-$ and $\mathbf{K}_k^c$ as follows:

$$\mathbf{P}_k = (\mathbf{I}_{6 \times 6} - \mathbf{K}_k^c \mathbf{H}^c) \mathbf{P}_k^-. \quad (20)$$

3.4 Autonomous landing controller

The controller for autonomous landing using the estimated relative position is designed as shown in Fig. 2(b). To direct the camera installed in front of the UAV toward the target, the desired yaw angular velocity $\omega_{z,\text{des}}$ is set as follows:

$$\omega_{z,\text{des}} = K_{p,\omega_z} \left\| \begin{bmatrix} \cos \psi \\ \sin \psi \\ 0 \end{bmatrix} \times \begin{bmatrix} (\mathbf{p}_{\text{target}}^{\text{uav}})_{xy} \\ 0 \end{bmatrix} \right\| \quad (21)$$

where K_{p,ω_z} is the proportional gain, ψ denotes yaw angle of the UAV, and $(\mathbf{p}_{\text{target}}^{\text{uav}})_{xy}$ denotes a vector of X and Y-axes components of $\mathbf{p}_{\text{target}}^{\text{uav}}$, which is estimated through the designed EKF. Similarly, the desired position $\mathbf{p}_{\text{des}}^w$ is set as follows:

$$\mathbf{p}_{\text{des}}^w = \mathbf{p}_{\text{target}}^w + \cot \theta_c (\mathbf{p}_{\text{target}}^{\text{uav}})_z \begin{bmatrix} \cos \psi \\ \sin \psi \\ 0 \end{bmatrix} \quad (22)$$

where θ_c is the angle at which the camera is tilted downward and $(\mathbf{p}_{\text{target}}^{\text{uav}})_z$ is Z-axis component of $\mathbf{p}_{\text{target}}^{\text{uav}}$. With the second term in Eq. (22), the autonomous landing controller controls the UAV in such a way that the moving platform can be located within the camera's FoV better. Then, the position error $\mathbf{e}$ is calculated as follows:

$$\begin{aligned} \mathbf{e} &= \mathbf{p}_{\text{des}}^w - \mathbf{p}_{\text{uav}}^w \\ &= \mathbf{p}_{\text{target}}^{\text{uav}} + \cot \theta_c (\mathbf{p}_{\text{target}}^{\text{uav}})_z \begin{bmatrix} \cos \psi \\ \sin \psi \\ 0 \end{bmatrix}. \end{aligned} \quad (23)$$

The proportional-integral-differential (PID) controller for horizontal velocity is constructed as follows:

$$\mathbf{V}_{xy,\text{des}} = K_p \mathbf{e}_{xy} + K_d \dot{\mathbf{e}}_{xy} + K_i \int \mathbf{e}_{xy} dt + \mathbf{V}_{\text{target}}^E \quad (24)$$

where $\mathbf{V}_{\text{target}}^E$ denotes the estimated velocity of the moving platform via encoder. Here, K denote PID gains corresponding to the subscripts. Similarly, in tracking mode, $\mathbf{V}_{z,\text{des}}$ is computed as follows:

$$\mathbf{V}_{z,\text{des}} = K_p (\mathbf{e}_z + h_{\text{fix}}) + K_d \dot{\mathbf{e}}_z + K_i \int (\mathbf{e}_z + h_{\text{fix}}) dt \quad (25)$$

Eq. (25) controls the UAV to follow the target while maintaining a fixed altitude h_{fix} from the target in tracking mode where the UAV is not close enough horizontally with the target to land yet. In landing mode, $\mathbf{V}_{z,\text{des}}$ is calculated as follows:

$$\mathbf{V}_{z,\text{des}} = -\mathbf{V}_{z,\text{max}} \frac{\mathcal{T}_{xy} - \|\mathbf{e}_{xy}\|}{\mathcal{T}_{xy}} \quad (26)$$

where $\mathbf{V}_{z,\text{max}}$ is the maximum descent speed and $\mathcal{T}_{xy}$ is the maximum horizontal error. In this way, the UAV descends only when $\|\mathbf{e}_{xy}\|$ is less than $\mathcal{T}_{xy}$, and as $\mathbf{e}_{xy}$ increases, it descends more slowly, to guarantee the precise and stable landing on the moving platform.

4. EXPERIMENTS AND RESULTS

4.1 Experiment Setup

To test the performance of the proposed algorithm, ROLAND, Gazebo [18] simulation is adopted with customized map¹, which is opened for public usages and has a bumpy ground. In the simulation, encoder data and UWB data can be obtained at most 20Hz and 10Hz when within the communicable range, respectively. VIO and object detection can operate at most 10Hz and 50Hz, respectively. To estimate the state of the UAV fairly, VINS-Stereo [15], was selected, which is one of the state-of-the-art and open-source methods and also was analyzed as one of the most precise algorithm in the author's early works [11], [12]. YOLOv4-tiny-3l network [17] is selected as object detector thanks to its lightweight network and low computational complexity considering the UAV's limited resources. UWB Gazebo plugin [14] is attached on the moving platform and the UAV. The experimental parameters are represented in Table 1.

Table 1. The experimental parameters

Parameter	Value	Parameter	Value
Camera FoV	90×67.5 [°]	$(\mathcal{T}_{xy}, \mathcal{T}_z)$	(0.25, 0.4) [m]
Camera depth range	6.0 [m]	$\mathcal{T}_z$	0.03 [m/s]
UWB range	10.0 [m]	$\mathcal{T}_{\text{ground}}$	0.1 [m]
$\mathcal{T}_{\text{YOLO}}$	0.8	h_{fix}	2.0 [m]

In the simulation, the UAV took off first and the moving platform started to run away at 0.8 m/s along the rectangular trajectory. During the first 50 seconds, the UAV only track the moving platform in tracking mode (not landing on it) to see the performance of the designed EKF's relative pose estimation. After 50 seconds, the UAV tried to land on the moving platform.

¹<https://github.com/engcang/gazebo.maps>

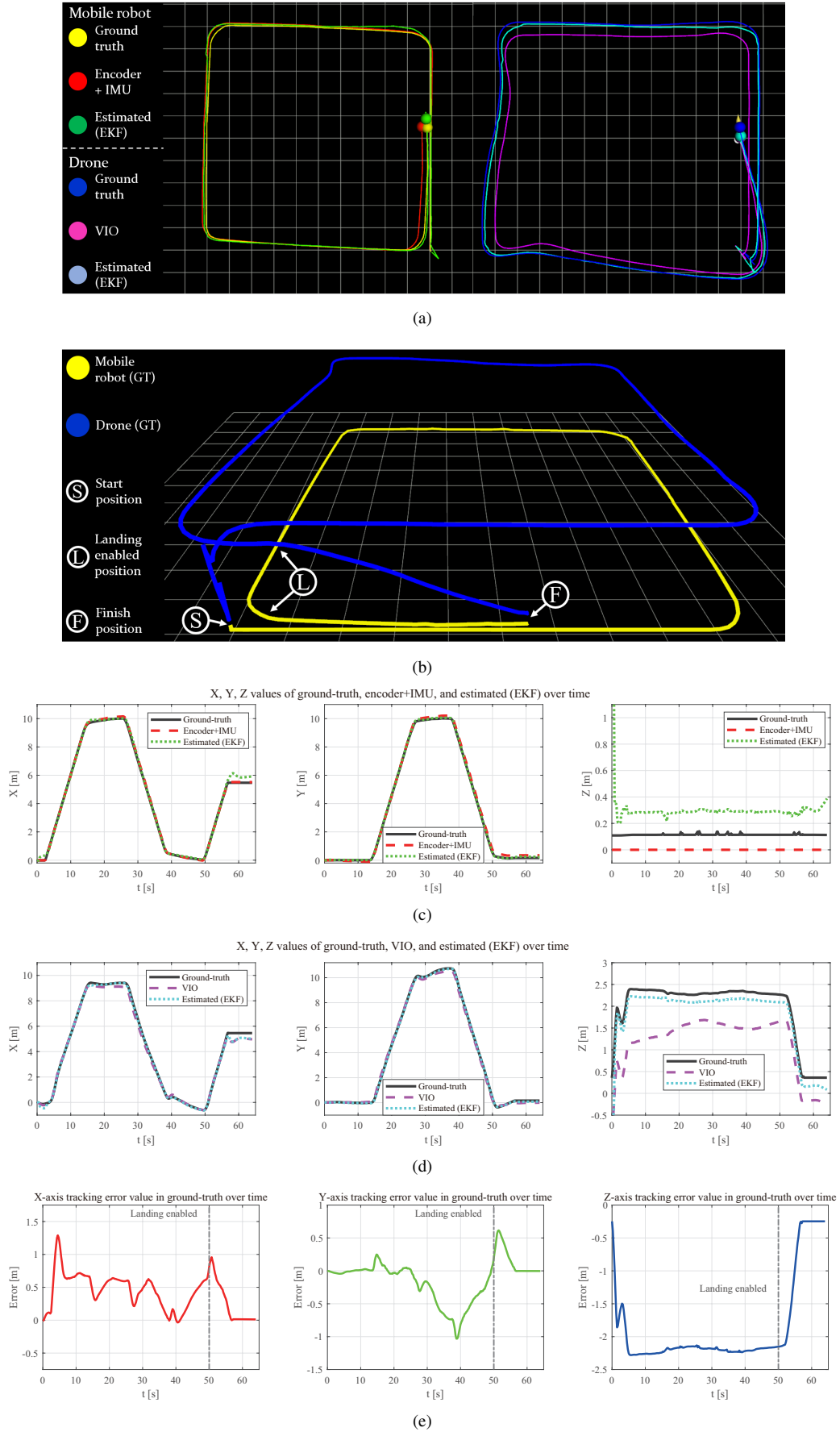


Fig. 4. The results of the simulation. (a), (b) The trajectories of the UAV and the moving platform. (c)–(e) The estimated states compared with the ground truth in X, Y, and Z-axis values. (c) The moving platform's ones which were estimated by the encoder with IMU or the proposed EKF. (d) The UAV's ones which were estimated by VIO or the proposed EKF. (e) The tracking error of the proposed controller. The landing mode was enabled after 50 seconds.

4.2 Results

The quantitative and qualitative results of the simulation can be seen in the Fig. 4. During the first 50 seconds, the UAV only tracked the moving platform while estimating the relative pose with the VIO, encoder, UWB and object detection data. As shown in Fig. 4(a), (c), and (d), the proposed method robustly estimated the states. Thanks to UWB, encoder, and VIO, when the target was outside the FoV of the camera, EKF was continuously updated and propagated. In addition, as can be seen in Fig. 4(e), the autonomous landing controller successfully tracked the moving platform and hence the UAV landed on it within fifteen seconds after landing mode started.

5. CONCLUSION

In this paper, we proposed a novel robust landing system of the UAV on moving platform using object detection and UWB based EKF. The proposed system, ROLAND, robustly estimated relative pose of the UAV and the moving platform and thus the UAV could successfully land on the target. The real-world experiment is left as the future works with the augmented state vector including the relative heading from the UAV to the target. The video of our simulation can be viewed at https://youtube.com/playlist?list=PLvgPHeVm_WqLP50wNDHvI9gPrta9TcGVT.

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